

JACUP, Australia

WMO: 956360

Lat: 33.886S

Lon: 119.111E

Elev: 306

StdP: 97.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	3.5	-2.3	3.2	20.2	-0.4	3.8	15.6	11.5	13.7	10.7	13.9	3.1	0	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.5	34.8	18.4	31.9	17.9	29.4	17.7	21.4	27.5	19.9	26.6	18.9	25.5	5.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	15.1	23.3	18.2	13.6	20.9	17.1	12.6	19.7	63.6	26.8	58.2	26.8	54.8	25.5	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.5	8.5	0.7	40.5	0.8	2.1	0.1	42.0	-0.4	43.2	-0.8	44.4	-1.4	45.9		
			0.4	26.3	0.7	4.4	-0.2	29.5	-0.6	32.0	-1.0	34.5	-1.5	37.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	20.0	20.0	18.6	16.4	13.6	11.3	10.2	10.7	12.1	14.3	16.8	18.4
	DBStd	4.59	3.41	3.24	3.65	3.20	2.64	2.07	2.01	2.26	2.82	3.35	3.66	3.37
	HDD10.0	63	0	0	0	0	2	9	22	17	10	3	0	0
	HDD18.3	1432	18	16	41	77	148	211	251	236	188	133	72	42
	CDD10.0	1952	309	280	266	191	114	48	29	41	72	136	206	259
	CDD18.3	278	69	63	48	18	2	0	0	0	1	8	27	42
	CDH23.3	2952	714	573	438	164	22	1	0	3	30	146	359	503
	CDH26.7	1254	333	251	191	50	3	0	0	6	49	151	221	
Wind	WSAvg	4.7	5.4	5.3	4.8	4.1	4.2	4.5	4.7	4.5	4.6	4.6	5.0	5.1
Precipitation	PrecAvg	415	26	19	32	34	35	44	44	46	40	34	33	26
	PrecMax	594	120	73	88	84	80	88	74	110	87	76	100	71
	PrecMin	237	0	1	0	3	12	21	15	15	7	4	1	0
	PrecStd	93	38	19	25	20	20	21	16	22	20	23	24	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	36.8	36.0	31.5	26.2	21.3	19.6	22.9	27.1	31.8	35.2	37.0
		MCWB	20.7	18.9	18.2	17.6	15.3	12.4	11.6	13.7	15.9	16.6	17.3	17.7
	2%	DB	33.9	33.1	31.7	27.6	23.0	19.0	17.5	19.4	23.0	27.4	31.0	32.1
		MCWB	19.4	18.7	17.6	16.6	14.0	12.0	11.6	12.6	14.0	15.5	17.0	17.1
	5%	DB	30.1	29.6	28.1	24.7	20.9	17.4	16.0	17.4	20.4	24.1	27.3	28.7
		MCWB	18.9	18.6	17.2	15.8	13.5	11.7	11.2	12.2	13.0	14.7	16.1	17.2
	10%	DB	27.0	26.6	24.9	22.3	19.0	15.8	14.7	15.7	18.2	21.3	24.2	25.5
		MCWB	17.7	17.7	16.5	15.4	13.0	11.3	10.8	11.5	12.3	13.7	15.2	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	23.6	20.9	20.4	17.5	14.9	14.0	16.7	18.6	19.3	20.8	20.9
		MCDB	30.6	28.6	26.5	24.2	21.6	17.8	16.3	17.7	20.8	25.2	28.1	29.3
	2%	WB	21.3	20.8	19.5	18.3	15.8	13.6	12.8	14.3	15.1	16.6	18.1	18.8
		MCDB	27.9	28.2	25.9	23.2	19.6	16.6	15.3	17.1	19.9	23.1	27.3	27.1
	5%	WB	19.6	19.4	18.5	17.2	14.8	12.8	11.9	13.0	13.7	15.3	16.8	17.8
		MCDB	27.0	25.6	24.3	21.7	18.5	15.5	14.8	16.1	18.5	22.0	24.3	25.6
	10%	WB	18.4	18.5	17.6	16.2	13.9	12.0	11.1	11.9	12.7	14.2	15.7	16.8
		MCDB	25.4	24.4	22.8	20.1	17.2	14.5	13.9	14.9	17.2	20.0	22.3	23.6
Mean Daily Temperature Range	5% DB	MDBR	13.3	12.5	12.0	11.3	10.4	9.3	9.0	10.2	11.8	13.0	13.5	13.3
		MCDBR	19.2	18.2	17.1	14.7	13.2	11.5	10.8	12.7	15.5	17.8	19.0	19.0
		MCWBR	6.7	5.9	5.6	5.9	6.0	6.1	6.1	7.0	7.5	7.2	6.9	6.4
	5% WB	MCDBR	16.2	15.1	13.3	12.0	10.1	9.5	9.1	10.7	13.2	15.0	16.4	16.4
		MCWBR	7.5	6.2	5.3	6.5	5.7	6.0	5.8	7.0	7.7	7.2	6.9	6.6
Clear-Sky Solar Irradiance	taub	0.341	0.334	0.320	0.307	0.295	0.288	0.286	0.292	0.316	0.319	0.326	0.329	
	taud	2.530	2.562	2.592	2.624	2.623	2.618	2.618	2.594	2.502	2.525	2.532	2.544	
	Ebn at Noon	997	983	962	921	881	861	882	923	950	987	1005	1012	
	Edh at Noon	111	103	93	80	71	67	70	81	100	106	110	110	
All-Sky Solar Radiation	RadAvg	7.49	6.57	5.37	3.98	2.98	2.50	2.69	3.52	4.71	5.98	6.96	7.68	
	RadStd	0.53	0.43	0.33	0.24	0.19	0.11	0.13	0.14	0.17	0.33	0.43	0.46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	N/A

Nomenclature: See separate page